
p-adic Representation Theory

— *EYNTKA* Series —

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0.0.1 Inspiration Very heavily inspired by [2]
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For a review of p -adic integers and p -adic fields (more generally of local fields), see [5, chapter 1.1].

1

Smooth Representations

There is a problem in my intuition here because we do *complex* representations!

Recall that if $\text{Gal}(K^{\text{ab}}/K)$ is the maximal abelian extension of K , it is isomorphic to the profinite completion of the idele group $C_K = \mathbb{I}_K/K^*$. When $K = \mathbb{Q}$, this becomes the restricted product $\prod'_p \mathbb{Z}_p^*$ showing us that there are natural maps $\text{Gal}(K^{\text{ab}}/K) \rightarrow \mathbb{Q}_p^* = \text{GL}_1(\mathbb{Q}_p)$; i.e. one-dimensional representations. When $\text{Gal}(L/K)$ is non-abelian, we get higher-dimensional (irreducible) representations:

$$\text{Gal}(L/K) \rightarrow \text{GL}_n(\mathbb{Q}_p)$$

As $\text{Gal}(\overline{K}/K)$ represents all finite extensions of K it is often more desirable to look for representation:

$$\text{Gal}(\overline{K}/K) \rightarrow \text{GL}_n(\mathbb{Q}_p)$$

In this section, we explore the basics of such representations. Just like for Lie groups, we will need to define the equivalent of “continuous” and “smooth” representations or else there would be too many representations. We then outline the natural generalizations.

1.1 Locally Profinite Groups

All finite groups will by assumption have the discrete topology. For a deeper review of topological groups, see [4, chapter 8].

The field $\mathbb{Q}_p$ is a *locally profinite groups* with its topology endowed by the metric $|\cdot|_p$. This is a substantially different topology to $\mathbb{C}$, and hence a moment must be taken to understand better its structure.

Definition 1.1.1: P-adic Topology

Let $\mathbb{Q}_p$ be a p-adic field for a (finite) prime p and let $\mathbb{Z}_p$ be the corresponding ring of integers. Define the neighbourhood basis:

$$\mathbb{Z}_p \supseteq p\mathbb{Z}_p \supseteq p^2\mathbb{Z}_p \supseteq \dots$$

Giving $\mathbb{Z}_p$ a p -adic topology which can be easily extended to $\mathbb{Q}_p$. As each $p^k\mathbb{Z}_p$ is a closed ball, this matches the metric topology induced by $|\cdot|_p$:

$$B_r(a) = \{x \in \mathbb{Q}_p \mid |x - a|_p < r\}$$

which extends to $\mathbb{Q}_p$ by defining:

$$\left| \frac{m}{n} \right|_p = \frac{|m|_p}{|n|_p}$$

Let us establish some basic properties of this field.

Lemma 1.1.2: P-adic Field Totally Disconnected

The field $\mathbb{Q}_p$ with the topology induced by the metric $|\cdot|_p$ is totally disconnected. More generally any space X whose topology is induced by an ultra-metric is totally disconnected.

Proof:

Let $a, b \in \mathbb{Q}_p$. As $a \neq b$, there must exist a d such that:

$$0 < |a - b|_p = p^{-k} = d$$

Let:

$$B_d(a) = \{x \in \mathbb{Q}_p \mid |x - a|_p < d\}$$

By construction, $a \in B_d(a)$ and $b \notin B_d(a)$.

Let us show that

$$B_d(a)^c = \{x \in \mathbb{Q}_p \mid |x - a|_p \geq d\}$$

is also open. Take $x \in B_d(a)^c$ so that $D = |a - x|_p \geq d$ and consider $B_d(x)$ and take $y \in B_d(x)$ so that $\delta = |x - y|_p \leq d$. We want to show that $y \in B_d(a)^c$ so $|y - a|_p \geq d$. Then by the isosceles property of the ultra-metric, for any three points a, x, y two of $d(a, x)$, $d(a, y)$ or $d(x, y)$ must be to the larger one. As $|a - x|_p \geq d$, $|y - a| \geq d$, and so $y \in B_d(a)^c$, showing that $B_d(x) \subseteq B_d(a)^c$ showing that it is *also* open.

Thus, $\mathbb{Q}_p$ is totally disconnected, as we sought to show.

Lemma 1.1.3: P-adic Integers Open Subgroup

The subgroup $\mathbb{Z}_p \subseteq \mathbb{Q}_p$ is an open subgroup with the topology induced by the valuation. Hence, it is also closed

This differs strongly from $\mathbb{C}$: if $H \subseteq \mathbb{C}$ is an open subgroup then as noted earlier it must also be

closed, hence connected, and hence $H = \mathbb{C}$. As $\mathbb{Q}_p$ is totally disconnected, it is possible to have clopen subsets that are not the entire set.

Proof:

First, $\mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid |x|_p \leq 1\}$. Take $a \in \mathbb{Z}_p$ (so that $|a| \leq 1$) and consider:

$$B_1(a) = \{x \in \mathbb{Q}_p \mid |x - a|_p < 1\}$$

I claim $x \in \mathbb{Z}_p$. Indeed:

$$|x|_p = |x + (-a + a)|_p \leq \max(|x - a|_p, |a|_p) = \max(t, 1) = 1$$

where $t < 1$ as $|x - a|_p < 1$. But then $x \in \mathbb{Z}_p$, and hence $B_1(a) \subseteq \mathbb{Z}_p$.

Thus, as $\mathbb{Q}_p$ is a topological group, $\mathbb{Z}_p$ is also closed.

There shall be instances where we work with other similar groups constructed from p -adic fields, for example finite extension of $\mathbb{Q}_p$, the ring of integers of $L_{\mathfrak{p}}/\mathbb{Q}_p$, $\overline{\mathbb{Q}_p}$, $\mathbb{C}_p$, $\mathrm{GL}_n(\mathbb{Q}_p)$ or $\mathrm{GL}_n(\mathcal{O}_{L/\mathbb{Q}_p})$ (or matrix subgroups of this group), $\mathrm{GL}_n(\mathfrak{m}_{\mathfrak{p}})$ and so forth. Thus, it is worth going over a general theory that encapsulates the topology of these spaces. Let us first recall some basic properties of topological groups:

- The topology of G is translation invariant: If $U \subseteq G$ is open, then Ux and xU is open
- If $H \leq G$, G/H has the quotient topology, and if $H \trianglelefteq G$ then G/H is a topological group.
- If K_1, K_2 are compact subset of G , so is $K_1 K_2$
- subgroups in relation to open/closed:
 - if $H \leq G$ is open, it must be closed and hence connected ($\bigcup_{g \in G \setminus H} gH$ is the union of open sets and the compliment of H). More strictly, if H is open if and only if it contains an open neighbourhood around the identity (if $e \in U \subseteq H$ with open U , then $hU \subseteq H$ and $H = \bigcup_{h \in H} hU$). If G is connected, it has no non-trivial proper open subgroups. We shall frequently be working with totally disconnected groups.
 - A closed subgroup $H \leq G$ need not be open (ex. $\mu_n \subseteq \mathbb{C}$). If G is first-countable, H is closed if it contains all its limit points (more generally, it contains all the limits of nets).
 - A subgroup need not be open or closed (ex. $\mathbb{Q} \subseteq \mathbb{R}$).
 - if H is a subgroup of G , so is $\overline{H}$ (the closure under limits/nets).
- symmetric neighbourhood:
 - For every open neighborhood of e , say for example U , there is a symmetric neighborhood (if $A \subseteq G$ such that $A = A^{-1}$, we'll say A is *symmetric*) V of e such that $V \subseteq U$
 - For every neighborhood U of e , there is a neighborhood V of e with $VV \subseteq U$

Lemma 1.1.4: Characterizing Hausdorffness

Let G be a topological group, $e \in G$ the identity, and $H = \bigcap_{\substack{U \subseteq G \text{ open} \\ e \in U}} U$. Then:

1. H is a subgroup
2. H is the closure of e , i.e. $\bar{e} = H$
3. G/H is Hausdorff
4. G is Hausdorff if and only if $H = \{e\}$

Proof:

exercise

Henceforth, we shall always assume Hausdorffness unless specified otherwise.

Definition 1.1.5: Profinite Group

Take a diagram finite groups with the discrete topology. Then a *profinite group* is a topological group that is the colimit of the system. Given any topological group G , we can take its *profinite completion* by taking:

$$\widehat{G} = \varprojlim_N G/N$$

where N ranges over all finite index open normal subgroups ordered by containment.

Any group G can be given the profinite topology by taking the cosets of finite-index normal subgroups as a basis. The reader should check that if G is a topological group and $\widehat{G}$ is its profinite completion, then $\varphi : G \rightarrow \widehat{G}$ is a continuous morphism and G is dense in $\widehat{G}$.

Example 1.1.6: Profinite Groups

We have the following “trivial” examples:

1. Let G be finite. Then all normal subgroups have finite index and there are finitely many. It follows that $G \cong \widehat{G}$.
2. Take $G = \mathbb{Q}$. Then $\mathbb{Q}$ has only one finite-index normal subgroup, namely itself. Hence $\widehat{\mathbb{Q}} = \{1\}$.

The following is the “usual” example and is better suited for the intuition of profinite groups. Let $G = \mathbb{Z}$. Then

$$\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}_p$$

where $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ is the p -adic completion of $\mathbb{Z}$. The main idea is that for any element $n \in \mathbb{Z}$, in chain of normal subgroups (say $p^k\mathbb{Z} \subseteq p^{k+1}\mathbb{Z} \subseteq \dots$) the element mod $p^k\mathbb{Z}$ eventually stabilize. For example, take $p = 7$ and $n = 129'654 = 2 \cdot 3^3 \cdot 7^4$. Then the value is zero for the until $\mathbb{Z}/7^5\mathbb{Z}$ where it becomes $12'005 \bmod 7^n$ for $n \in \{5, 6\}$ and then $129'654$ forever onward. $\mathbb{Z}_p$ allows for never-stabilizing chains (mirroring how stable sequences of rational numbers converge to rational numbers).

For Langlands, the important non-abelian example is the profinite completion of galois Galois groups, namely $\text{Gal}_K(\overline{K})$. See [6] or a quick overview or [1] for more details.

More generally, profinite groups can be seen as the object that captures all the finite behaviour “up to infinity” (or philosophically, all the “potential infinity”) of a system. In the $\mathbb{Z}_p$ example, it appeared when taking a solution mod each p^n and for galois it comes from amalgamating all symmetries of roots of some infinite family of (finite) polynomial. Another place this shall show up is in the definition of the étale fundamental group, see [ref:HERE](#).

The above examples show that the map $\varphi : G \rightarrow \widehat{G}$ could be injective, surjective, both.

Lemma 1.1.7: Pro-finite Completion Injective

Let G be a topological group and $\widehat{G}$ its profinite completion. Then $G \rightarrow \widehat{G}$ is injective if the intersection of all finite-index open normal subgroups of G is trivial

Proof:
exercise.

The completion comes equipped with a natural universal property:

Theorem 1.1.8: Universal Property of Profinite Completion

let $\varphi : G \rightarrow H$ be continuous homomorphism with H a profinite group. Then there is a unique continuous homomorphism such that the following diagram commutes:

$$\begin{array}{ccc} G & \xrightarrow{\phi} & \widehat{G} \\ & \searrow \varphi & \downarrow \exists! \\ & & H \end{array}$$

Proof:
exercise (hint: recall G is dense in $\widehat{G}$).

A striking result is that all profinite groups can be characterized by a simple topological property:

Theorem 1.1.9: Characterizing Profinite Groups

A topological group is profinite if and only if it is a totally disconnected compact group

Proof:
Let G be profinite. Then it is a closed subgroup of a direct product of finite groups, hence compact and totally disconnected.
Conversely, suppose G is totally disconnected and compact. We must show it comes from an inverse system of finite groups. [will finish later](#).

Corollary 1.1.10: Distinguishing Profinite Completion

Let G be a profinite group. Then G is naturally isomorphic to its profinite completion, or even stronger:

$$G \cong \varprojlim G/U$$

where U ranges over all *open* normal subgroups ordered by contain need.

If G is isomorphic to its profinite completion *as a group* (i.e. is strongly complete) if and only if every finite index subgroup is open

Proof:
exercise.

Definition 1.1.11: Locally Profinite Group

A topological group G is *locally profinite* if it is Hausdorff and every open neighbourhood of the identity contains a compact open subgroup of G .

Notice how there is no mention of an inverse system. The reader should show that the definition implies G is totally disconnected, and hence if it is also compact then we recover a profinite group. Every discrete group is finite, closed subgroups of a locally profinite group are locally profinite, and so are quotients by closed normal subgroups. A locally profinite subgroup is locally compact. It is an exercise to show that any locally compact totally disconnected group is locally profinite. The canonical example for these notes of a locally profinite group is the non-Archimedean local field $(F, +)$ as proven in [5]. Similarly, $(F^\times, \times)$ is a profinite group by passing through e^x , or:

$$U_F^n = 1 + \mathfrak{p}^n$$

Being profinite is closed under products, hence F^n is profinite, and $(M_n(F), +)$ is locally profinite under addition. The group $\mathrm{GL}_n(F) \subseteq M_n(F)$ under multiplication is an open topological subgroup. It is locally profinite as $\mathrm{GL}_n(\mathcal{O}_F)$ is profinite (it is compact as it is homeomorphic to a closed subset of $\mathcal{O}_F^{n^2}$ which is a product of compact groups, and it is open as $\mathcal{O}_F = \{x \in F \mid |x|_{\mathfrak{p}} \leq 1\}$ is an open set. Note that this was all done in coordinates, more generally this applies to $\mathrm{End}_F(V)$ and $\mathrm{Aut}_F(V)$.

Let us now consider character's on locally profinite groups:

Lemma 1.1.12: Continuous Character

let $\varphi : G \rightarrow \mathbb{C}^\times$ be a group homomorphism. Then the following are equivalent:

1. φ is continuous
2. the kernel of φ is open

If φ satisfies one of these conditions and G is the union of its compact open subgroups, then $\mathrm{im} \varphi \subseteq S^1 \subseteq \mathbb{C}^\times$.

Proof:

If the kernel is open, then if $U \subseteq \mathbb{C}^\times$ and $K = \ker \varphi$,

$$\varphi^{-1}(U) = \bigcup_{g \in \varphi^{-1}(U)} gK$$

showing φ is continuous. Conversely, Let φ be continuous and U an open neighbourhood of 1. Then $\varphi^{-1}(U)$ is open, and thus by local-profiniteness contains a compact open subgroup $K \subseteq G$. As this applies to any U , we can pick U sufficiently small so that it contains no non-trivial subgroup of $\mathbb{C}^\times$. As the image of group is a group and $K \subseteq \varphi^{-1}(U)$, $K \subseteq \ker \varphi$. As K is open, $\ker \varphi$ contains an open subgroup and hence by topological group theory $\ker \varphi$ is open.

Next, S^1 is the unique maximal compact subgroup of $\mathbb{C}^\times$. If K' is a compact subgroup of G , $\varphi(K') \subseteq S^1$, giving the 2nd result as we sought to show.

Definition 1.1.13: Continuous Character

A *character* for a locally profinite group G is a continuous homomorphism

$$\varphi : G \rightarrow \mathbb{C}^\times$$

We shall often write 1_G or just 1 for the trivial character (i.e. the constant character).

If $\varphi(G) \subseteq S^1$, φ is called a *unitary character*.

summary of next section On page 12 in [2], he defines $\widehat{F}$ to be the set of all characters of F . As F is a local field, it is the union of its compact open subgroup $\mathfrak{p}^n$ for $n \in \mathbb{N}$, hence all characters are unitary. Then if $\varphi \neq 1$ is a character, there exists $d \in \mathbb{Z}$ such that $\mathfrak{p}^d \subseteq \ker(\varphi)$; the least such number is defined as the level of φ . Fixing a d , the set of characters of level $\leq d$ is a subgroup. This defines a duality which extends to the multiplicative set, in other words this is indeed the right notion of character!

1.2 Smooth Representations of Locally Profinite Groups

All representations shall be complex unless specified otherwise. When working with $\mathbb{Q}_p$ representations and more generally representations of locally profinite groups, it is important to take into account the topology, or else we will get representations that give no valuable information:

Example 1.2.1: “Useless” Representation

Take the algebraic closure of $\mathbb{Q}_p$, $\overline{\mathbb{Q}_p}$. By field theory, it embeds into $\mathbb{C}$ (using the axiom of choice). Thus there is an injective ring homomorphism $\mathbb{Q}_p \rightarrow \mathbb{C}$ and hence a (valid) representation:

$$\pi : \mathrm{GL}_n(\mathbb{Q}_p) \rightarrow \mathrm{GL}_n(\mathbb{C})$$

But this has disregarded all topological (and hence arithmetic) information of $\mathbb{Q}_p$.

The first step to fix this would be to only consider continuous representations as is done in the

representation of topological groups. Often when G is not compact, we get a vector space V that is infinite dimensional with which is either a Banach or Hilbert space and $\mathrm{GL}(V)$ is the set of all invertible bounded linear operators; this thread is further explored in [3].

When restricting to locally profinite groups, there is a useful stronger notion called *smooth*. It must be that this notion of smoothness is different from smoothness for manifolds as we will be looking at maps $\mathbb{Q}_p \rightarrow \mathbb{C}$ which don't share the local topology ($\mathbb{Q}_p$ is also totally disconnected, so there is no way to put a reasonable atlas on it). Instead, the intuition comes from how smoothness (or differentiability) means that zooming in close will get the same linear, or “constant” if we change coordinates in our perspective. In terms of locally profinite groups, as they are the union of open compact subgroups, a representation being constant on these groups abstractly shares this same property. This loose analogy turns out to be very practical, hence:

Definition 1.2.2: Smooth Map For Locally Profinite Groups

Let G be a locally profinite group. Then a map $f : G \rightarrow \mathbb{C}^n$ is *smooth* if it is locally constant, namely for any $x \in G$ there exists an open neighbourhood U containing x such that $f(x) = f(y)$ for all $x, y \in U$.

1.2.3 Precision in Vocabulary Via Harmonic Analysis There is a more precise comparison when considering the Fourier transform with distributions, so we'll assume [4] for the following.

1. On $\mathbb{R}$: For the Fourier transform, the space of “ideal” functions is the Schwartz space, consisting of functions that are both smooth (C^∞) and rapidly decaying. The Fourier transform is a perfect map on this space: the transform of a smooth, rapidly decaying function is another smooth, rapidly decaying function.
2. On $\mathbb{Q}_p$: There is an analogous theory of Fourier analysis. The “ideal” space of functions (called the Schwartz-Bruhat space) consists of functions that are locally constant and have compact support. The Fourier transform on $\mathbb{Q}_p$ is a good map on this space: the transform of a locally constant, compactly supported function is another locally constant, compactly supported function. See [this link](#) for more details.

Proposition 1.2.4: Equivalence Notion of Smoothness

Let G be a locally profinite group and $f : G \rightarrow \mathbb{C}^n$ a map. Then f is smooth if and only if for any subset $U \subseteq \mathbb{C}^n$ (which is *not* necessarily open), $f^{-1}(U) \subseteq G$ is open.

Thus, if f is smooth, it is continuous.

Proof:
exercise.

The key in the notion of smoothness being useful is that on $\mathbb{Q}_p$ there are non-trivial locally constant functions:

Example 1.2.5: Smooth Functions On Locally Profinite Groups

1. The p -adic valuation $|\cdot|_p : \mathbb{Q}_p^\times \rightarrow \mathbb{R}$ is smooth. To see this, take any $x \in \mathbb{Q}_p$ so that $|x|_p = p^{-n}$ for some $n \in \mathbb{Z}$ giving the range. Then the pre-image of p^{-n} is $p^n \mathbb{Z}_p^\times$ which is open.

Note that this is not smooth on $\mathbb{Q}_p$ because the pre-image of 0 is 0, i.e. there is no open neighbourhood of 0 on which the absolute value is constant. This can even be stretched to the absolute value on $\mathbb{R}$, when restricting to $\mathbb{R}^\times$ it is smooth, but naturally it is not smooth at 0.

2. The map $\mathrm{GL}_n(\mathbb{Q}_p) \rightarrow \mathbb{R}$ given by $g \mapsto |\det(g)|_p$ is smooth.

By identifying $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$, we get the notion of a smooth representation:

Definition 1.2.6: Smooth Representation

Let G be a locally profinite group. Then a *representation* of G is a pair (π, V) where V is a complex vector space and $\pi : G \rightarrow \mathrm{Aut}_{\mathbb{C}}(V)$ a group homomorphism.

A representation (π, V) is called *smooth* if for every $v \in V$, there is a compact open subgroup $K_v \subseteq G$ such that for all $k \in K_v$:

$$\pi(k) \cdot v = v$$

Equivalently, if V^K denotes the space of $\pi(K)$ -fixed vector in V ,

$$V = \bigcup_K V^K$$

where K ranges over all compact open subgroups of G .

A smooth representation is called *admissible* if V^K is finite dimensional for all compact open $K \subseteq G$.

Proposition 1.2.7: Smooth Representation Alternative

A representation $\pi : G \rightarrow \mathrm{GL}(V)$ is smooth if

$$\mathrm{Stab}_G(v) = \{g \in G \mid \pi(g) \cdot v = v\}$$

is an open subgroup of G for all $v \in V$

Proof:
exercise.

If (π, V) is a smooth representation of G , then any G -stable subspace of V is another smooth representation, and similarly if $U \subseteq V$ is a G -subspace, the induced representation of G on V/U is smooth. A representation (π, V) is *irreducible* if $V \neq 0$ and V has no nontrivial G -stable subspaces. A G -homomorphism between two representations $(\pi_1, V_1), (\pi_2, V_2)$ is a map that commutes with the

group action induce by π_i .

Example 1.2.8: Smooth Representations

1. A continuous character $\chi : G \rightarrow \mathbb{C}^\times = \text{Aut}_{\mathbb{C}}(\mathbb{C})$ is a smooth representation $(\chi, \mathbb{C})$. Note that a 1-dimensional representation of G is smooth if and only if it is equivalent to a representation defined by a character. Indeed, if $\chi(g) \cdot v = v$, as $\chi(g)$ is a complex number and v is nonzero we have:

$$\chi(g) = 1$$

Hence, χ is smooth if and only if the kernel is open (and as G is locally profinite it contains a compact subgroup), which by lemma 1.1.12 is true if and only if χ is continuous.

2. Let G be compact (thus profinite) and (π, V) an irreducible representation of G . Then V must be compact. Indeed, if $0 \neq v \in V$, then $v \in V^K$ for an open subgroup $K \subseteq G$. As G is profinite, $[G : K]$ is finite and so:

$$\{\pi(g) \cdot v \mid g \in G/K\}$$

spans V . We can then take the smaller $K' = \bigcup_{g \in G/K} gKg^{-1}$ which is an open normal subgroup of G , still of finite index. Then V can be seen as an irreducible representation of the finite discrete group G/K' , and hence is certainly finite dimensional.

With morphisms defined, we shall denote the category of smooth representations of G , $\text{Rep}(G)$; by a similar argument as the usual category for (not necessarily smooth) representations of finite groups, $\text{Rep}(G)$ is abelian an abelian category. When working with finite groups, we saw that representations can be thought of as $\mathbb{C}[G]$ -modules. With the smoothness condition imposed, there are more elements in the $\mathbb{C}[G]$ -module V than smooth representations; we shall need to limit to *Hecke algebras* as shall be discussed in [ref:HERE](#).

Let us next upgrade the standard decomposition result from finite group representation theory:

Proposition 1.2.9: Semisimple Equivalence

Let G be a locally profinite group, and let (π, V) be a smooth representation of G . The following conditions are equivalent:

1. V is the sum of its irreducible G -subspaces;
2. V is the direct sum of a family of irreducible G -subspaces;
3. any G -subspace of V has a G -complement in V .

Proof:

(1) $\Rightarrow$ (2). Take the set of irreducible G -subspaces U_i of V

$$\{U_i : i \in I\}$$

such that $V = \sum_{i \in I} U_i$. Consider the set $\mathcal{I}$ of subsets J of I such that the sum $\sum_{i \in J} U_i$ is direct. The set $\mathcal{I}$ is non-empty; Indeed, for an induction argument suppose we have a totally ordered set $\{J_a : a \in A\}$ of elements of $\mathcal{I}$. Put $J = \bigcup_{a \in A} J_a$. If the sum $\sum_{j \in J} U_j$ is not direct, there is

a finite subset S of J for which $\sum_{j \in S} U_j$ is not direct. Since S must be contained in some J_a , this is impossible. Therefore $J \in \mathcal{I}$. We can now apply Zorn's Lemma to get a maximal element J_0 of $\mathcal{I}$. For this set, we have

$$V = \bigoplus_{a \in J_0} U_i,$$

as required for (2).

(2) $\Rightarrow$ (3). In (3), let W be a G -subspace of V . By (2), we can assume that $V = \bigoplus_{i \in I} U_i$, for a family (U_i) of irreducible G -subspaces of V . We consider the set $\mathcal{J}$ of subsets J of I for which $W \cap \sum_{i \in J} U_i = 0$. Again, the set $\mathcal{J}$ is nonempty and inductively ordered by inclusion. If J is a maximal element of $\mathcal{J}$, the sum $X = W + \bigoplus_{j \in J} U_j$ is direct. If $X \neq V$, there is $i \in I$ with $U_i \not\subset X$, so the sum $X + U_i$ is direct, and $J \cup \{i\} \in \mathcal{J}$, contrary to hypothesis. Thus (2) $\Rightarrow$ (3).

(3) $\Rightarrow$ (1) Let V_0 be the sum of all irreducible G -subspaces of V and write $V = V_0 \oplus W$, for some G -subspace W of V . Assume for a contradiction that $W \neq 0$. By its definition, the space W has no irreducible G -subspace. However, there is a non-zero G -subspace W_1 of W which is finitely generated over G . By Zorn's Lemma, W_1 has a maximal G -subspace W_0 , and then W_1/W_0 is irreducible. We have $V = V_0 \oplus W_0 \oplus U$, for some G -subspace U of V , and hence a G -projection $V \rightarrow U$. The image of W_1 in U is an irreducible G -subspace of U , hence an irreducible subspace of V which is not contained in V_0 . This is nonsense, so $V = V_0$ and (3) $\Rightarrow$ (1).

Definition 1.2.10: Semisimple Representation

A representation (π, V) is *G -semisimple* if it satisfies one of the equivalent conditions of proposition 1.2.9.

As we will see, locally profinite groups will have many non-semisimple representations. however, their compact opens shall fair better:

Lemma 1.2.11: Compact Open Semistable Representation

Let G be a locally profinite group, and let K be a compact open subgroup of G . Let (π, V) be a smooth representation of G . The space V is the sum of its irreducible K -subspaces.

Thus V is K -semisimple.

Proof:

Let $v \in V$. As in example 1.2.8, v is fixed by an open normal subgroup K' of K , and it generates a finite-dimensional K -space W on which K' acts trivially. Thus W is effectively a finite-dimensional representation of the finite group K/K' and so is the sum of its irreducible K -subspaces. Since $v \in V$ was chosen at random, the lemma follows.

Thus, the following is a meaningful definition:

Definition 1.2.12: Isotypic Component

Let G be a locally profinite group and $K \subseteq G$ a compact open subgroup. Let $\widehat{K}$ denote the set of equivalence classes of irreducible representations. Let $\rho \in \widehat{K}$ and (π, V) be a smooth representation of G . Then:

$$V^\rho = \sum_{\substack{W \subseteq V \\ K\text{-subspace} \\ \text{of class } \rho}} W$$

is called the ρ -isotypic component of V .

By definition, V^K is the isotypic subspace for the class of trivial representation 1 of K .

Proposition 1.2.13: K -isotypic Decomposition

Let (π, V) be a smooth representation of G and let K be a compact open subgroup of G .

1. The space V is the direct sum of its K -isotypic components:

$$V = \bigoplus_{\rho \in \widehat{K}} V^\rho.$$

2. Let (σ, W) be a smooth representation of G . For any G -homomorphism $f : V \rightarrow W$ and $\rho \in \widehat{K}$, we have

$$f(V^\rho) \subset W^\rho \text{ and } W^\rho \cap f(V) = f(V^\rho).$$

Proof:

Proof. We use 2.2 to write

$$V = \bigoplus_{i \in I} U_i,$$

for a family of irreducible K -subspaces U_i of V . We let $U(\rho)$ be the sum of those U_i of class ρ . We then have

$$V = \bigoplus_{\rho \in \widehat{K}} U(\rho).$$

If W is an irreducible K -subspace of V of class ρ , then $W \subset U(\rho)$; otherwise, there would be a non-zero K -homomorphism $W \rightarrow U_i$, for some U_i of some class $\tau \neq \rho$. We deduce that $V^\rho = U(\rho)$ and (1) follows.

In (2), the image of V^ρ is a sum of irreducible K -subspaces of W , all of class ρ and therefore contained in W^ρ . Moreover, $f(V)$ is the sum of the images $f(V^\tau)$, $\tau \in \widehat{K}$, and $f(V^\tau) \subset W^\tau$. Since the sum of the W^τ is direct, $f(V)$ is the direct sum of the $f(V^\tau)$ and the second assertion follows.

We frequently use part (2) of the Proposition in the following context:

Corollary 1.2.14: Environment Title

Corollary 1. Let $a : U \rightarrow V$, $b : V \rightarrow W$ be G -homomorphisms between smooth representations U, V, W of G . The sequence

$$U \xrightarrow{a} V \xrightarrow{b} W$$

is exact if and only if

$$U^K \xrightarrow{a} V^K \xrightarrow{b} W^K$$

is exact, for every compact open subgroup K of G .

Proof:
exercise.

If H is a subgroup of G , we define

$$V(H) = \text{the linear span of } \{v - \pi(h)v : v \in V, h \in H\}. \quad (2.3.1)$$

In particular, $V(H)$ is an H -subspace of V .

Corollary 1.2.15: Environment Title

Corollary 2. Let G be a locally profinite group, and let (π, V) be a smooth representation of G . Let K be a compact open subgroup of G . Then

$$V(K) = \bigoplus_{\substack{\rho \in \hat{K} \\ \rho \neq 1}} V^\rho, \quad V = V^K \oplus V(K),$$

and $V(K)$ is the unique K -complement of V^K in V .

Proof:

Proof. The sum $W = \bigoplus V^\rho$, with ρ not trivial, is a K -complement of V^K in V . There is therefore a K -surjection $V \rightarrow V^K$ with kernel W . Clearly, $V(K)$ is contained in the kernel of any K -homomorphism $V \rightarrow V^K$; we conclude that W contains $V(K)$. On the other hand, if U is an irreducible K -space of class $\rho \neq 1$, then $U(K) = U$, so $V^\rho \subset V(K)$.

The next important concept is that of induced representation and how we can construct representations using them

Definition 1.2.16: Induced Representation

Let G be a locally profinite group, and let H be a closed subgroup of G so H is also locally profinite. Let (σ, W) be a smooth representation of H . Consider the space X of functions $f : G \rightarrow W$ which satisfy:

1. $f(hg) = \sigma(h)f(g)$, $h \in H, g \in G$;
2. there is a compact open subgroup K of G (depending on f) such that $f(gx) = f(g)$, for $g \in G, x \in K$

Then define a homomorphism $\Sigma : G \rightarrow \text{Aut}_{\mathbb{C}}(X)$ by

$$\Sigma(g)f : x \mapsto f(xg), \quad g, x \in G.$$

The pair (Σ, X) provides a smooth representation of G and is said to be *smoothly induced by* σ . It is usually denoted

$$(\Sigma, X) = \text{Ind}_H^G \sigma.$$

Proposition 1.2.17: Induced Representation is A Functor

The map $\sigma \mapsto \text{Ind}_H^G \sigma$ gives a functor $\text{Rep}(H) \rightarrow \text{Rep}(G)$.

Proof:

exercise

There is a canonical H -homomorphism

$$\alpha_\sigma : \text{Ind}_H^G \sigma \longrightarrow W, \quad f \mapsto f(1).$$

The pair $(\text{Ind}_H^G \sigma, \alpha)$ has the following fundamental property:

Theorem 1.2.18: Frobenius Reciprocity

Let H be a closed subgroup of a locally profinite group G . For a smooth representation (σ, W) of H and a smooth representation (π, V) of G , the canonical map

$$\begin{aligned} \text{Hom}_G(\pi, \text{Ind}_H^G \sigma) &\longrightarrow \text{Hom}_H(\pi, \sigma) \\ \varphi &\mapsto \alpha_\sigma \circ \varphi, \end{aligned}$$

is an isomorphism that is functorial in both variables π, σ .

Proof:

Let $f : V \rightarrow W$ be an H -homomorphism. We define a G -homomorphism $f_* : V \rightarrow \text{Ind} \sigma$ by letting $f_*(v)$ be the function $g \mapsto f(\pi(g)v)$. The map $f \mapsto f_*$ is then the inverse of (2.4.2).

The following shows how this is a very useful functor:

Proposition 1.2.19: Induced Representation is Additive and Exact

The functor $\text{Ind}_H^G : \text{Rep}(H) \rightarrow \text{Rep}(G)$ is additive and exact.

Proof:

For a smooth representation (σ, W) of H , temporarily let $I(\sigma)$ denote the space of functions $G \rightarrow W$ satisfying the first condition $f(hg) = \sigma(h)f(g)$ of the definition above. Thus I is a functor to the category of abstract representations of G ; it is clearly additive and exact, while $\text{Ind}_H^G(\sigma) = I(\sigma)^\infty$. Thus Ind_H^G is surely additive, and 2.3 Exercise (3) shows it to be left-exact.

To prove it is right-exact, let $(\sigma, W), (\tau, U)$ be smooth representations of H and let $f : W \rightarrow U$ be an H -surjection. Take $\varphi \in I(\tau)^\infty$, and choose a compact open subgroup K of G which fixes φ . The support of φ is a union of cosets HgK , and the value $\varphi(g) \in U$ must be fixed by $\tau(H \cap gKg^{-1})$. By 2.3 Corollary 1 (applied to the group H and the trivial representation of its compact open subgroup $H \cap gKg^{-1}$), there exists $w_g \in W$, fixed by $\sigma(H \cap gKg^{-1})$, such that $f(w_g) = \varphi(g)$. We define a function $\Phi : G \rightarrow W$ to have the same support as φ and $\Phi(hgk) = \sigma(h)w_g$, for each $g \in H \backslash \text{supp} \varphi / K$. The function Φ is fixed by K , and hence lies in $I(\sigma)^\infty$. Its image in $I(\tau)^\infty$ is φ , as required.

A lot more here, including importantly a generalization of Shur's Lemma

1.3 Measures and Duality

Here

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